

## 1 Introduction

The goal of this project is to develop and evaluate a 3D object detection model capable of accurately and efficiently perceiving the surrounding environment for autonomous vehicles. This includes detecting and localizing objects in three-dimensional space, which is fundamental for tasks such as safe navigation, obstacle avoidance, and decision-making in autonomous driving systems. Our focus is on analyzing 2D images captured by monocular cameras.

To this end, we experimented with two state-of-the-art camera-based 3D object detection models—SparseBEV and StreamPETR—conducting both pretrained model inference and fine-tuning on various datasets, to assess how different models impact detection quality across varied scenarios. By comparing these results, this study aims to evaluate the feasibility of camera-based 3D perception as a reliable backup or complement to LiDAR.

## 2 Background

3D object detection plays a fundamental role in autonomous driving, enabling the perception system to detect and localize surrounding objects in 3D space. Historically, most solutions rely on LiDAR sensors due to their high precision, but recent research is exploring whether camera-based models can achieve comparable performance at lower cost. In this work, we review several camera-based 3D detection models and concentrate on two leading approaches, SparseBEV and StreamPETR, as potential LiDAR backup solutions. These models were selected for their reported superior performance on Nuscenes.

### 2.1 LiDAR-based Methods

LiDAR-based detection models provide accurate depth information and are robust across a variety of

driving conditions. These models operate on point clouds, using either voxel-based or point-based architectures to extract spatial features. While effective, they come with several drawbacks: high sensor costs, limited resolution at long range, and difficulties in detecting small objects.

### 2.2 Camera-based Approaches

Camera-based methods are generally more affordable compared to LiDAR. While LiDAR remains a widely used and reliable technology, camera-based perception leverages semantic understanding to detect objects that are often more difficult for LiDAR to identify. Despite the challenge of depth ambiguity, state-of-the-art camera-only models have achieved encouraging results, especially when exploiting BEV (Bird’s Eye View) representations or temporal sequences. BEV is a representation that projects the scene into a top-down view, making it easier to reason about spatial layouts.

### 2.3 SOA Models

State-of-the-art 3D object detection models rely on either LiDAR or camera inputs, each offering distinct advantages and trade-offs. LiDAR provides precise geometry but is costly, while cameras offer rich visual cues at lower expense. This section reviews representative approaches from both paradigms, emphasizing their main contributions.

- **LiDAR-based methods:**

Representative LiDAR-based approaches include **M3DeTR**[5], which addresses the limitations of voxel-based representations such as PV-RCNN++ by leveraging an attention mechanism to reduce the semantic gap across different embedding spaces. Trained on both the KITTI and Waymo Open datasets, the unified architecture achieves state-of-the-art

performance. The model takes raw point clouds as input and combines point, voxel, and BEV-based features into shared embedding spaces. Keypoint features extracted through a point-voxel set abstraction module are then passed to the M3 transformer.

Another state-of-the-art model is **FocalFormer3D** [1], addresses the challenge of missed detections (false negatives) in 3D object detection, particularly for difficult instances. Instead of using a conventional global self-attention mechanism, the paper introduces Hard Instance Probing (HIP), a general pipeline that identifies and focuses on hard-to-detect instances in a multi-stage manner. This approach guides the model to improve recall and excavate difficult objects. The model then uses a box-level transformer decoder to efficiently distinguish objects from a large number of candidates. Additionally, FocalFormer3D leverages a multi-stage heatmap encoder to generate high-recall object candidates and aggregates multi-scale features, allowing it to balance detailed local information with broader contextual cues. This design leads to faster inference and stronger detection performance, as validated by experimental results on the nuScenes and Waymo datasets.

- **Camera-based methods:**

Representative camera-based approaches include **MonoCInIS**[2], a camera-independent method that employs instance segmentation as the core detection mechanism. Unlike depth-based monocular 3D detectors, it does not rely on explicit depth estimation, which makes it adaptable to datasets collected with different camera models by instead leveraging 2D reference points. The model has been trained on a variety of large-scale benchmarks, including KITTI-3D, nuScenes, and Waymo.

**SparseBEV** [7] is a fully sparse 3D object detector that avoids the expensive construction of dense BEV feature maps and the reliance on heavy encoder architectures such as FPNs or stacked transformers. Instead, the method reformulates the encoder by using self-attention directly among BEV queries, allow-

ing them to interact and exchange contextual information without passing through intermediate dense grids. To further enrich the representation, SparseBEV introduces scale-adaptive self-attention, enabling each query to dynamically adjust its receptive field in order to capture both local fine-grained details and broader multi-scale context. In addition, the model integrates adaptive spatio-temporal sampling to align the most informative regions across frames, ensuring temporal coherence, and employs adaptive mixing in the decoder to selectively combine these features in an efficient way. By combining these components, SparseBEV achieves state-of-the-art accuracy while maintaining efficiency, making it suitable for real-time 3D perception tasks.

**StreamPETR** [6] extends the PETR framework to streaming video by introducing an *object-centric temporal modeling* paradigm. Instead of relying on dense BEV features, it propagates long-term information through object queries stored in a memory queue and updated via a propagation transformer. To further enhance temporal reasoning, a motion-aware layer normalization explicitly incorporates ego-motion and object dynamics, allowing the model to better capture both the movement of the vehicle and the trajectories of surrounding agents. This design enables efficient online 3D detection with minimal latency, making the method particularly suitable for real-time applications such as autonomous driving. By effectively leveraging temporal consistency and reducing redundant computation, StreamPETR achieves state-of-the-art performance on nuScenes and significantly narrows the performance gap with LiDAR-based methods, demonstrating the potential of camera-only systems for reliable perception in safety-critical scenarios.

### 3 Datasets

In this section we provide details on the data we have used for the work. In particular, there are three main datasets: Nuscenes, a test set, used for testing scope, and a training set, that has been exploited for fine-tuning purposes. The latter two, as

described after, have been manually annotated, for simplicity we will refer to them as Dataset AIDA. All the datasets contain monocular driving images.

### 3.1 Nuscenes

The NuScenes dataset has been employed as an additional benchmark to evaluate the generalization capabilities of our approach. It is a large-scale autonomous driving dataset that provides multimodal sensor data, including images from multiple cameras, LiDAR, and radar. For our work, we focused on the monocular camera stream to ensure consistency with the other datasets. NuScenes offers a wide range of driving scenarios with diverse weather and lighting conditions, which makes it a suitable reference for validating the robustness of our models.

## 3.2 AIDA Dataset

### 3.2.1 Annotation of the Dataset

Regarding the annotation of the dataset, the test set was manually annotated using the SUSTech annotation tool [3]. Since labelling the Training Validation manually set has been very time consuming, in order to scale the labelling procedure, pseudo-labels were generated using BEVFusion [4], a Multimodal Camera-LiDAR model, and were then manually refined to reduce false positives and improve label quality. The images were subsequently resized to match the input dimensions required by the models.

### 3.2.2 Training/Validation Set

We then prepared a second dataset of approximately 1000 images for training and validation usage. The split was made of 80% training set and the remaining 20% to the validation set. As showed in 1, also for this dataset we have a prevalence of cars, followed by the presence of pedestrians, and then with other categories including trucks, motorcycles, bicycles and bus.

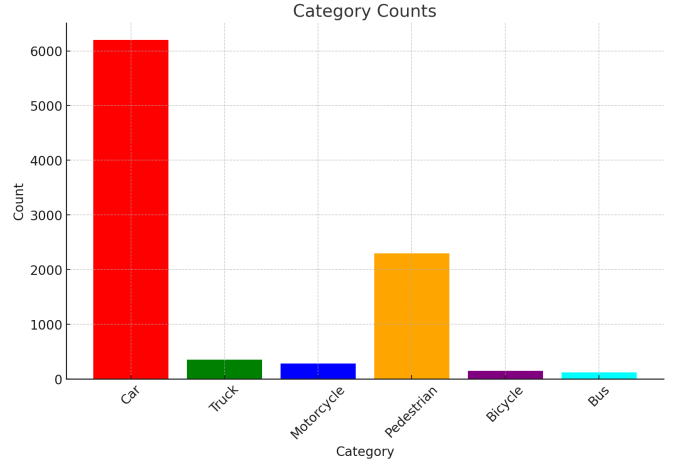


Figure 1: Class distribution of the training set

### 3.2.3 Test set

The first dataset has been used for testing our final model. It is composed of approximately 300 images, and it was provided without annotations. In order to perform the evaluation of the performance on this dataset a labelling process have been done. Given the high time consuming task of labelling, we used the SUSTech annotation tool [3]. As we can see from 2, the dataset is predominantly composed of car instances, with the presence also of pedestrians and bicycle.

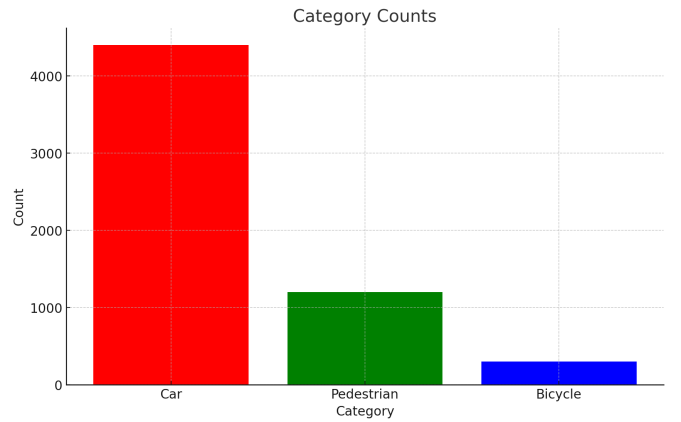


Figure 2: Class distribution of the test set

## 4 Methodology

Building on the context outlined above, this section provides an overview of how the study was conducted. It includes the description of the experimental setups for the two selected models, outlining

the steps taken to adapt them to the aims of our study.

#### 4.1 SparseBEV: Training and Evaluation Options

The SparseBEV model was evaluated under several different training conditions to assess its adaptability and performance:

- **Training on nuScenes:** the model was firstly used without any additional fine-tuning to perform inference on both nuScenes and our custom validation dataset. The weights were obtained by the authors, by training on the nuScenes dataset. Then, the model was re-trained from scratch on nuScenes, modifying the input by removing all additional sweeps and keeping only one keyframe for each sample. This change reduced the input from 8 frames to 1.
- **Fine-tuning on AIDA dataset:** we performed additional fine-tuning on the model initialized with the weights obtained from the previous step. We have used the following configuration: 5 epochs, the backbone was freezed, an adaptable learning rate starting from 1e-5 and we've set the learning rate of the FPN to 0 to avoid to update their weights.

#### 4.2 StreamPETR: Training and Evaluation Options

StreamPETR was evaluated under two different setups:

- **Training on nuScenes:** run inference on our validation set with pretrained weights(that the authors obtained by training on the nuScenes dataset) to assess baseline performance and generalization to our domain.
- **Fine-tuning on AIDA dataset:** starting from pretrained weights, we fine-tuned on our pseudo-labeled images (train-val dataset) to better adapt to our specific environment and camera setup. We use the following configuration: 3 epochs, backbone we unfreeze 1 layer, adaptable learning rate starting from 1e-4, and set the learning multiplier of the neck and roi to 0 to avoid them being trained.

Given its superior results compared to SparseBEV on AIDA's validation set, only StreamPETR was tested on our AIDA test set.

## 5 Results

Following the methodology described earlier, this section presents the main findings of our experiments. It compares the performance of the two models across different training settings using the nuScenes Detection Score (NDS), which combines measures of detection accuracy and localization into a single score, and the mean Average Precision (mAP), as the primary comparison metrics. We have slightly adapted the way the NDS is computed, since for our use case the velocity error and the attribute error are not relevant. In 1 the updated formula for the NDS can be found.

$$\begin{aligned}
 NDS = & 0.5 * mAP \\
 & + 0.2 * mATE \\
 & + 0.15 * mASE \\
 & + 0.15 * mAOE
 \end{aligned} \tag{1}$$

The analysis highlights trends, improvements, and limitations observed throughout the evaluation process.

#### 5.1 Performance on nuScenes Dataset

| StreamPETR |         |       |       |            |            |       |
|------------|---------|-------|-------|------------|------------|-------|
|            | Bicycle | Bus   | Car   | Motorcycle | Pedestrian | Truck |
| AP         | 0.516   | 0.647 | 0.741 | 0.579      | 0.611      | 0.763 |
| ATE        | 0.600   | 0.640 | 0.355 | 0.547      | 0.605      | 0.302 |
| ASE        | 0.447   | 0.062 | 0.087 | 0.336      | 0.242      | 0.057 |
| AOE        | 0.412   | 0.812 | 0.116 | 0.654      | 0.311      | 0.070 |

| SparseBEV |         |       |       |            |            |       |
|-----------|---------|-------|-------|------------|------------|-------|
|           | Bicycle | Bus   | Car   | Motorcycle | Pedestrian | Truck |
| AP        | 0.297   | 0.600 | 0.707 | 0.580      | 0.653      | 0.590 |
| ATE       | 0.662   | 0.661 | 0.390 | 0.586      | 0.501      | 0.393 |
| ASE       | 0.193   | 0.091 | 0.155 | 0.281      | 0.268      | 0.156 |
| AOE       | 0.453   | 0.063 | 0.097 | 1.116      | 0.421      | 0.042 |

| FocalFormer3D |         |       |       |            |            |       |
|---------------|---------|-------|-------|------------|------------|-------|
|               | Bicycle | Bus   | Car   | Motorcycle | Pedestrian | Truck |
| AP            | 0.496   | 0.696 | 0.872 | 0.762      | 0.882      | 0.570 |
| ATE           | 0.186   | 0.273 | 0.165 | 0.180      | 0.127      | 0.304 |
| ASE           | 0.265   | 0.169 | 0.145 | 0.226      | 0.297      | 0.175 |
| AOE           | 0.406   | 0.038 | 0.074 | 0.239      | 0.316      | 0.057 |

| Model                           | mAP    | NDS    |
|---------------------------------|--------|--------|
| StreamPETR pre-trained          | 0.4576 | 0.4940 |
| SparseBEV pretrained            | 0.4315 | 0.4746 |
| FocalFormer3D (from scoreboard) | 0.687  | 0.726  |

The comparison between StreamPETR and SparseBEV using pretrained weights on the nuScenes dataset highlights key differences in their detection capabilities. StreamPETR demonstrates overall similar performance. It shows a consistent object detection across dynamic classes such as vehicles and pedestrians, likely benefiting from its spatiotemporal feature aggregation mechanism.

SparseBEV exhibits competitive results in terms of shape estimation and achieves comparable performance in mAP and NDS. However, it suffers from higher orientation errors, which may stem from its reduced reliance on temporal cues.

For comparison we have also added the result of a Lidar-based model - FocalFormer3D - which are visible better than the ones obtained so far.

These findings suggest that both models are viable out-of-the-box for monocular 3D detection tasks, resulting in similar performance over nuScenes dataset.

## 5.2 Performance on Dataset AIDA

### 5.2.1 Validation Dataset

**SparseBev on Dataset AIDA** (around 200 frames)

|                                               | Bicycle | Bus   | Car   | Motorcycle | Pedestrian | Truck |
|-----------------------------------------------|---------|-------|-------|------------|------------|-------|
| <i>Pretrained</i>                             |         |       |       |            |            |       |
| AP                                            | 0.012   | 0.182 | 0.725 | 0.033      | 0.044      | 0.325 |
| ATE                                           | 0.81    | 0.88  | 1.33  | 0.71       | 0.36       | 1.18  |
| ASE                                           | 0.24    | 0.22  | 0.13  | 0.18       | 0.21       | 0.21  |
| AOE                                           | 1.22    | 0.91  | 0.41  | 0.19       | 1.42       | 0.08  |
| <i>Trained from Scratch</i>                   |         |       |       |            |            |       |
| AP                                            | 0.017   | 0.164 | 0.603 | 0.210      | 0.067      | 0.360 |
| ATE                                           | 0.80    | 0.88  | 1.01  | 0.65       | 0.33       | 1.11  |
| ASE                                           | 0.22    | 0.24  | 0.16  | 0.17       | 0.18       | 0.33  |
| AOE                                           | 2.07    | 1.58  | 0.21  | 0.88       | 1.34       | 0.10  |
| <i>Fine-Tuned (from Trained from Scratch)</i> |         |       |       |            |            |       |
| AP                                            | 0.000   | 0.182 | 0.118 | 0.000      | 0.006      | 0.109 |
| ATE                                           | –       | 0.74  | 1.71  | –          | 0.38       | 1.29  |
| ASE                                           | –       | 0.17  | 0.11  | –          | 0.14       | 0.37  |
| AOE                                           | –       | 0.91  | 1.35  | –          | 1.08       | 0.81  |

**StreamPETR on Dataset AIDA**

|                   | Bicycle | Bus   | Car   | Motorcycle | Pedestrian | Truck |
|-------------------|---------|-------|-------|------------|------------|-------|
| <i>Pretrained</i> |         |       |       |            |            |       |
| AP                | 0.000   | 0.002 | 0.132 | 0.000      | 0.000      | 0.005 |
| ATE               | –       | 1.86  | 2.39  | –          | 0.34       | 1.89  |
| ASE               | –       | 0.52  | 0.13  | –          | 0.35       | 0.36  |
| AOE               | –       | 2.97  | 0.44  | –          | 1.45       | 0.31  |
| <i>Fine-Tuned</i> |         |       |       |            |            |       |
| AP                | 0.000   | 0.428 | 0.518 | 0.227      | 0.050      | 0.346 |
| ATE               | –       | 0.74  | 1.53  | 0.46       | 0.32       | 1.06  |
| ASE               | –       | 0.49  | 0.15  | 0.15       | 0.12       | 0.17  |
| AOE               | –       | 0.17  | 0.34  | 0.25       | 1.61       | 0.12  |

| Model - SparseBev                      | mAP    | NDS   |
|----------------------------------------|--------|-------|
| Pretrained                             | 0.2201 | 0.299 |
| Trained from Scratch                   | 0.2367 | 0.272 |
| Fine-tuned (from Trained from Scratch) | 0.0690 | 0.143 |

| Model - Stream-PETR | mAP    | NDS     |
|---------------------|--------|---------|
| Pretrained          | 0.0233 | -0.3453 |
| Fine-tuned          | 0.2615 | 0.4810  |



Figure 3: Example detection output on nuScenes dataset with StreamPETR model.

Unlike the nuScenes dataset, our validation set revealed a performance gap between the two models, with StreamPETR outperforming SparseBEV

overall — especially after fine-tuning — thanks to its flexibility on our data.

- **SparseBev**

The results show a small improvement in detection accuracy when retraining SparseBEV from scratch using nuScenes keyframes only (i.e., with a single frame and no LiDAR sweeps). This setup achieved the highest mAP (0.2367) but exhibits a modest decline in NDS (0.272) compared to the original pretrained model. This suggests that adapting the model to a monocular-like setup improves slightly its compatibility with our data.

However, further fine-tuning on our pseudo-labeled dataset significantly reduced the mAP to 0.0690 and NDS to 0.143. As shown in the per-class results, fine-tuning degraded AP on all classes except bus, and in some cases, AP dropped to zero.

This trade-off underlines the importance of label quality and class balance in the fine-tuning dataset, and suggests that additional filtering or relabeling steps might be required before fine-tuning SparseBEV effectively.

- **StreamPETR**

The results show a substantial improvement in performance for the StreamPETR model after fine-tuning on our dataset. Notably, the mAP increased from 0.0233 to 0.2615, while the NDS improved from a negative value (-0.3453), which indicates poor detection quality, to a considerably stronger 0.4810. This demonstrates that fine-tuning allowed the model to better adapt to the characteristics of our target domain.

On a per-class basis, almost all relevant categories benefited from fine-tuning. For instance, the car class saw a rise in average precision (AP) from 0.132 to 0.518, and the bus class improved from 0.002 to 0.428. These gains indicate a more confident and consistent prediction behavior, especially for dominant object categories. The model also reduced the number of false positives and improved localization quality (as reflected in ATE, ASE, and AOE) across several classes.

Overall, the fine-tuned StreamPETR improves its pretrained counterpart in both aggregate metrics and object-level performance, confirming the effectiveness of our domain-specific training pipeline.

### 5.2.2 Test Dataset

**StreamPETR on AIDA-mdp-500 Dataset**

|                                    | Bicycle | Car   | Pedestrian |
|------------------------------------|---------|-------|------------|
| <i>Pretrained Pre-Fine-Tuning</i>  |         |       |            |
| AP                                 | 0.018   | 0.340 | 0.002      |
| ATE                                | 0.88    | 1.79  | 0.37       |
| ASE                                | 0.38    | 0.27  | 0.64       |
| AOE                                | 2.65    | 0.85  | 1.98       |
| AVE                                | 0.38    | 1.14  | 0.86       |
| AAE                                | 0.88    | 0.59  | 0.93       |
| <i>Fine-Tuned Post-Fine-Tuning</i> |         |       |            |
| AP                                 | 0.003   | 0.473 | 0.062      |
| ATE                                | 0.84    | 1.17  | 0.35       |
| ASE                                | 0.57    | 0.21  | 0.43       |
| AOE                                | 0.82    | 0.64  | 1.98       |
| AVE                                | 0.00    | 0.01  | 0.00       |
| AAE                                | 0.00    | 0.02  | 0.00       |

| Model      | mAP    | NDS    |
|------------|--------|--------|
| Pretrained | 0.1200 | -0.057 |
| Fine-tuned | 0.1790 | 0.359  |

As StreamPETR showed superior performance to SparseBEV on the validation set and demonstrated further improvements during fine-tuning, we evaluated it exclusively on the custom test set.

After fine-tuning StreamPETR on the AIDA MDP 500 dataset, we observed a substantial improvement in both class-wise and overall metrics, with a decision threshold set at 0.2. The most notable changes are observed in the *Average Precision (AP)* values across all evaluated classes. For instance, the AP for the *car* class increases from 0.340 to 0.473, indicating a substantial gain in detection performance. Similarly, *pedestrian* class show improvements, going from nearly undetectable (0.002) to 0.620 mAP.

In addition to AP, localization and orientation metrics such as *ATE*, *ASE*, and *AOE* show marked reductions, confirming that the model not only detects better but also localizes objects more precisely. For example, the ATE for the car class decreases from 1.79 to 1.17, and the AOE drops from 0.85 to 0.64. These improvements are crucial for applications requiring accurate positioning and orientation of detected objects, such as autonomous driving.

Overall, the fine-tuning process effectively adapts the model to the specific characteristics of

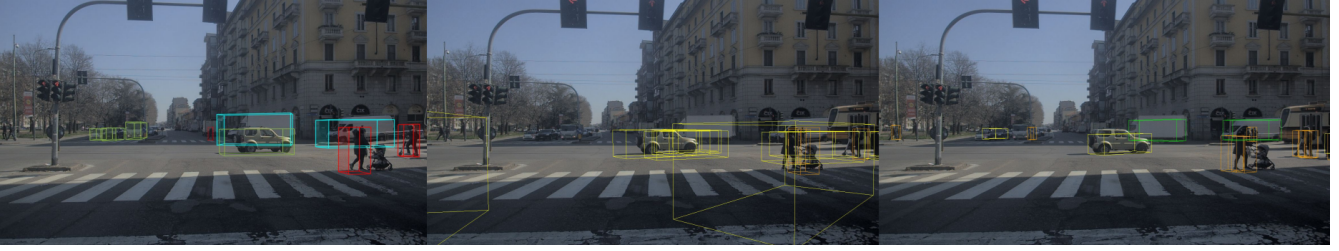


Figure 4: Example detection output from the front camera on the Training-Validation dataset, with bounding boxes from ground-truth labels (left), SparseBEV fine-tuned (center), and StreamPETR fine-tuned (right).

the AIDA MDP 500 dataset, leading to a considerable performance boost. These results highlight the importance of domain-specific fine-tuning when deploying pretrained 3D object detectors in new or specialized environments.



Figure 5: Example detection output from the front camera with bounding boxes (Frame 15) of aida\_500 dataset using StreamPETR model.

### 5.3 Limitations and Future Work

Future research could explore extended fine-tuning with longer training schedules and a larger number of images to improve generalization. Increasing dataset diversity—both in environmental conditions and camera setups—could help reduce overfitting and enhance model robustness. Additionally, ensuring a more balanced representation of different object classes, not just cars, would help create a more equitable dataset and improve detection performance across all categories.

## 6 Conclusion

In this project, we explored two state-of-the-art camera-based 3D object detection models—SparseBEV and StreamPETR—for their applicability in autonomous driving scenarios using

monocular images.

Our experiments show that, while both models can perform 3D detection without LiDAR, a LiDAR-based system still remains necessary given the performance gaps observed. SparseBEV demonstrated strong performance in pretrained inference on nuScenes and showed improvements when retrained on single-frame nuScenes data, evaluated on our custom test and validation dataset. However, fine-tuning on our custom dataset led to performance degradation, likely due to limited generalization capacity within the dataset.

On the validation dataset, StreamPETR achieved overall better performance than SparseBEV, with higher overall mAP and NDS scores. Fine-tuning it on our pseudo-labeled dataset significantly improved detection on our manually labeled AIDA-MDP-500 test set, making it a more stable and adaptable choice for varied scenarios.

Overall, our findings suggest that camera-only models can serve as a useful complement to LiDAR, but not yet a complete replacement. Future work will focus on improving adaptation to different camera characteristics, refining fine-tuning strategies, and integrating temporal consistency and uncertainty estimation to enhance performance and reliability.

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